

## ORIGINAL RESEARCH

## SGLT2 Inhibitors Promote Diabetic Wound Healing Via AMPK/AKT/mTORC1-Regulated Endothelial Angiogenesis

Furkan Bestepe<sup>1</sup>, George F. Ghanem, Caroline Kelly, Kamran Huseynli, Danielle V. Richards, Rachel Barber<sup>2</sup>, Annabelle Zamary, Parul Sahu<sup>3</sup>, Payam Salehi<sup>4</sup>, Gordon S. Huggins<sup>5</sup>, Basak Icli<sup>1</sup>

**BACKGROUND:** Approximately 25% of patients with type 2 diabetes are at risk of developing diabetic ulcers, which can progress to amputations. A key challenge in diabetic wound healing is impaired angiogenesis. Although SGLT2 (sodium-glucose transporter 2) inhibitors are known for their cardiovascular and renal benefits, their role in wound healing and angiogenesis, remains incompletely understood.

**METHODS:** We evaluated the effects of empagliflozin, dapagliflozin, and canagliflozin on wound healing in db/db mice by measuring wound closure, granulation tissue thickness, and perfusion. Angiogenesis and proliferation were assessed by CD31 and Ki67 immunostaining. Ex vivo studies used human skin organoids, sprouting assays from human microvessels and murine aortic explants. In vitro studies were performed in human umbilical vein endothelial cells (ECs).

**RESULTS:** Empagliflozin and dapagliflozin significantly improved wound healing, perfusion, angiogenesis, and cell proliferation while canagliflozin showed limited benefit. In human skin organoids and ECs, empagliflozin and dapagliflozin but not canagliflozin improved wound closure, angiogenesis, proliferation, and migration. Sprouting angiogenesis of human vascular explants and mice aorta showed significantly increased branching and junction formation in response to empagliflozin and dapagliflozin. All 3 SGLT2 inhibitors similarly suppressed inflammation and improved EC barrier function, while empagliflozin and dapagliflozin selectively increased tissue remodeling gene expression. Mechanistically, canagliflozin significantly increased AMPK (AMP-activated protein kinase) phosphorylation while decreasing AKT/mTORC1 (mechanistic target of rapamycin complex 1) activation compared with empagliflozin, dapagliflozin. AMPK inhibition with compound C, or dose reduction of canagliflozin, partially restored AKT/mTORC1 signaling and wound closure.

**CONCLUSIONS:** These findings uncover distinct angiogenic effects among SGLT2 inhibitors. Empagliflozin and dapagliflozin promote EC-angiogenic functions through balanced AMPK/AKT/mTORC1 signaling, whereas canagliflozin impaired these processes via excessive AMPK activation and suppression of AKT/mTORC1 signaling. AMPK inhibition or dose reduction restored AKT/mTORC1 signaling and partially rescued wound healing, underscoring the importance of balanced AMPK/AKT/mTORC1 signaling for EC-angiogenic function in diabetic wounds.

**Key Words:** AMP-activated protein kinases ■ angiogenesis ■ canagliflozin ■ dapagliflozin ■ empagliflozin ■ TOR serine-threonine kinases ■ wound healing

Type 2 diabetes (T2D) affects more than 10% of adults in the United States<sup>1</sup> and contributes to a wide range of vascular complications. Among these, diabetic foot ulcers (DFUs) and peripheral artery disease (PAD) are 2 conditions that significantly impair the

quality of life, contribute to disability, and lead to substantial health care costs.<sup>2</sup> DFUs affect ≈25% of people with T2D during their lifetime and are the leading cause of diabetes-related lower-limb amputations, accounting for 70% to 80% of nontraumatic lower limb amputations

Correspondence to: Basak Icli, PhD, Department of Medicine, Molecular Cardiology Research Institute, Tufts Medical Center, Boston, MA 02111. Email [bicli@tuftsmedicalcenter.org](mailto:bicli@tuftsmedicalcenter.org)

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## Nonstandard Abbreviations and Acronyms

|                                |                                                               |
|--------------------------------|---------------------------------------------------------------|
| <b>4E-BP1</b>                  | eukaryotic translation initiation factor 4E-binding protein 1 |
| <b>ACC</b>                     | acetyl-CoA carboxylase                                        |
| <b>AKT</b>                     | AKT serine/threonine kinase                                   |
| <b>AMPK</b>                    | AMP-activated protein kinase                                  |
| <b>CD31</b>                    | cluster of differentiation 31 (PECAM1)                        |
| <b>CompC</b>                   | compound C                                                    |
| <b>DFU</b>                     | diabetic foot ulcer                                           |
| <b>EC</b>                      | endothelial cell                                              |
| <b>HUVEC</b>                   | human umbilical vein endothelial cell                         |
| <b>mTORC1</b>                  | mechanistic target of rapamycin complex 1                     |
| <b>PAD</b>                     | peripheral artery disease                                     |
| <b>RPS6</b>                    | ribosomal protein S6                                          |
| <b>SGLT2</b>                   | sodium-glucose cotransporter 2                                |
| <b>T2D</b>                     | type 2 diabetes                                               |
| <b><math>\alpha</math>-SMA</b> | $\alpha$ -smooth muscle actin                                 |

## What Are the Clinical Implications?

Diabetic wound healing remains a major clinical challenge, contributing significantly to morbidity, limb amputation, and health care costs in patients with type 2 diabetes. Although SGLT2 (sodium-glucose cotransporter 2) inhibitors—empagliflozin, dapagliflozin, and canagliflozin—have demonstrated cardiovascular and renal benefits in large clinical trials, their direct effects on wound healing and vascular repair have remained largely unexplored. This study reveals important in-class differences among SGLT2 inhibitors in regulating diabetic wound healing. Local delivery of empagliflozin and dapagliflozin significantly improved wound closure, perfusion, and granulation tissue formation by promoting endothelial cell proliferation, migration, and angiogenesis. In contrast, canagliflozin showed limited wound healing benefit despite similar anti-inflammatory effects, an effect mechanistically linked to disproportionate AMPK (AMP-activated protein kinase) activation and consequent suppression of AKT (AKT serine/threonine kinase)/mTORC1 (mechanistic target of rapamycin complex 1) signaling critical for angiogenesis. These findings suggest that not all SGLT2 inhibitors are equivalent in the context of diabetic wound care, and that drug selection may meaningfully impact wound healing outcomes. Furthermore, the identification of AMPK/AKT/mTORC1 signaling balance as a key determinant of endothelial angiogenic function provides a novel therapeutic target for improving wound healing in diabetic patients.

in this population.<sup>2–4</sup> PAD, which affects nearly half of patients with DFUs, further impairs perfusion and healing, compounding the risk of limb loss.<sup>1,5</sup> These conditions are associated with significantly reduced quality of life and are a >2.5-fold increase in 5-year mortality among patients who undergo amputations.<sup>6</sup>

PAD and DFU frequently coexist and share overlapping pathophysiology, including impaired angiogenesis and vascular dysfunction.<sup>4,7,8</sup> Patients with both PAD and T2D have a significantly higher risk of developing a foot ulcer compared with those with diabetes alone.<sup>9</sup> These complications often coexist with other microvascular complications of diabetes such as neuropathy, nephropathy and retinopathy,<sup>10,11</sup> reflecting shared mechanisms of endothelial injury. DFUs remain difficult to treat, with recurrence rates around 20% that have not significantly improved over the last 20 years.<sup>12</sup> This underscores the limited progress in the field and highlights the scarcity of randomized trials evaluating the therapeutic efficacy of available treatment strategies.<sup>13,14</sup> Consequently, there is an urgent need for effective therapeutic approaches to treat diabetic wounds and prevent their progression to severe, untreatable stages. Given the central role of angiogenesis in tissue regeneration, and its impairment across multiple diabetic complications including DFU and PAD, strategies that restore vascular repair may hold broader therapeutic potential in the T2D population.

Angiogenesis is a coordinated multicellular process<sup>15</sup> in which endothelial cells (ECs) form the vascular scaffold required for wound repair. Adequate angiogenesis supports oxygen and nutrient delivery during tissue regeneration,<sup>16</sup> whereas T2D impairs this process and contributes to delayed wound healing.<sup>17</sup>

SGLT2 (sodium-glucose cotransporter 2) inhibitors have emerged as effective therapies for glycemic control in T2D<sup>18,19</sup> and also improve cardiac and renal outcomes beyond their glucose-lowering effects.<sup>20,21</sup> Emerging evidence suggests that SGLT2 inhibition improves endothelial function, vascular rarefaction, angiogenesis, and wound healing.<sup>22,23</sup> However, the mechanism by which they improve EC function remains unclear. Empagliflozin reduces reactive oxygen species generation and mitochondrial stress in hyperglycemic ECs,<sup>24</sup> whereas dapagliflozin improves angiogenic responses and diabetic wound healing through HIF-1 $\alpha$  (hypoxia-inducible factor-1 $\alpha$ ) signaling.<sup>25</sup> Conversely, SGLT2 inhibitors may exert divergent effects on EC biology; for example, canagliflozin inhibits EC proliferation and tube formation<sup>26</sup> and fails to enhance microvascular density in ischemic cardiac and skeletal muscle.<sup>27,28</sup> These findings highlight the complexity of SGLT2 inhibitor-mediated vascular responses.

SGLT2 inhibitors exert therapeutic effects beyond glycemic control, including attenuation of fibrosis across multiple organs. In clinical studies, SGLT2 inhibitor therapy has been associated with reductions in liver stiffness measurements, a noninvasive surrogate marker of hepatic fibrosis and steatosis, in patients receiving these medications.<sup>29</sup>

Additional investigations demonstrated reduced fibroblast activation and protective effects in human cardiac fibroblasts under high glucose stress.<sup>30</sup> Empagliflozin attenuates cardiac fibrosis in mice with T2D through modulation of TGF- $\beta$ /SMAD (transforming growth factor- $\beta$ /SMAD signaling pathway) signaling,<sup>31</sup> and dapagliflozin decreases lipopolysaccharide-induced myocardial fibrosis through AMPK (AMP-activated protein kinase) activation. These findings demonstrate the anti-fibrotic effects of SGLT2 inhibitors in cardiac and renal tissues.

SGLT2 inhibitors activate AMPK, a key energy sensor that promotes endothelial nitric oxide production, enhances EC survival, and supports angiogenesis under metabolic stress.<sup>32,33</sup> Conversely, they may suppress AKT (AKT serine/threonine kinase) signaling, which regulates EC proliferation, migration, and tube formation.<sup>34</sup> mTORC1 (mechanistic target of rapamycin complex 1) integrates nutrient and stress signals and maintains a reciprocal relationship with AMPK and AKT.<sup>35</sup> In nutrient-sensing contexts, AMPK inhibits anabolic processes, whereas mTORC1 promotes cellular growth.<sup>36</sup> The impact of SGLT2 inhibitors on mTORC1 appears context-dependent: suppression of excessive mTORC1 activity may confer cardioprotective benefits,<sup>34</sup> whereas complete inhibition may be detrimental during acute ischemia, where mTORC1 is required to sustain protein synthesis and limit infarct size.<sup>37</sup> These divergent effects on AMPK, AKT, and mTORC1 likely contribute to context-dependent vascular outcomes, reflecting a dynamic balance between AMPK-mediated survival signaling, AKT-driven proliferative responses, and mTORC1-regulated growth.

Herein, we investigated how local delivery of SGLT2 inhibitors influences wound healing dynamics in the microenvironment of diabetic cutaneous wounds, linking AMPK/AKT/mTORC1 signaling to vascular repair.

## METHODS

The [Major Resources Table](#), including catalog numbers and manufacturer information, is available in the [Supplemental Material](#). The authors declare that all supporting data are available within the article its [Supplemental Materials](#). Illustrations were created using BioRender.com.

## Ethical Approval

Experimental procedures were approved by the Tufts University Institutional Animal Care and Use Committee (IACUC [Institutional Animal Care and Use Committee] protocol number: B2023-123 [exp. date: August 3, 2027]) and conducted in accordance with the National Institutes of Health guidelines. Experiments with human samples were approved by the institutional review board (protocol number: 9487, STUDY00000793, [November 12, 2022–October 3, 2026]) at the Tufts Medical Center.

## Animal Studies

For dermal wound studies, 10- to 11-week-old, male homozygous db/db and heterozygous db/+ B6.BKS(D)-Leprdb/J

mice were used. Animals had ad libitum access to food and water. Blood glucose was measured by tail snapping. At the end point, mice were euthanized according to Tufts University IACUC guidelines. Tissues were snap-frozen for RNA/protein analyses or fixed in 10% formalin for histology. Blood was collected into EDTA (ethylenediaminetetraacetic acid) tubes and plasma was separated by 15 minutes, 1500× *g* centrifugation. Male mice were age-matched and cage-matched littermates were used. Analyses of in vivo samples were performed by blinded observers. All procedures were approved by the Laboratory Animal Care at Tufts Medical Center and complied with National Institutes of Health Guidelines.

## Experimental Murine Wound Healing Model

Murine dorsal skin wounding was performed as previously described.<sup>38</sup> Ten- to 11-week-old db/db or db/+ mice were anesthetized with isoflurane. Three full-thickness wounds were created with a 6-mm steel punch in triangle order on the dorsal area. Empagliflozin, dapagliflozin, canagliflozin, and 1% DMSO as vehicle control were intradermally delivered (1, 1, and 10 mg/kg, respectively) within 2 to 3 mm of the wound edges. For rescue studies, compound C (CompC; 1 mg/kg) was administered. Digital images were acquired on days 0, 2, 4, and 6 postwounding. After imaging, a semi-occlusive dressing (Tegaderm) was applied to the dorsum. Mice were euthanized 6 days postwounding, and tissue was harvested.

## SGLT2 Inhibitor and Glucose Dosing Rationale

Local administration of empagliflozin (1 mg/kg), dapagliflozin (1 mg/kg), and canagliflozin (10 mg/kg) was based on prior studies demonstrating effective wound neovascularization in db/db or STZ (streptozotocin)-induced diabetic mice. Previous work supports 10 mg/kg systemic or 1 to 10 mg/kg local dosing for empagliflozin<sup>39</sup> and dapagliflozin,<sup>25,40</sup> and 100 mg/kg systemic dosing for canagliflozin.<sup>41,42</sup> Systemic doses were converted to local doses as previously described,<sup>43,44</sup> with wound area (10–12 cm<sup>2</sup>) representing approximately one-tenth of the total-body surface area (100–120 cm<sup>2</sup>), allowing proportional dose adjustment for pharmacologically relevant tissue exposure. For in vitro and ex vivo studies, empagliflozin and dapagliflozin were used at 1  $\mu$ mol/L, and canagliflozin at 10  $\mu$ mol/L, based on reported peak plasma concentrations<sup>45,46</sup> and prior EC studies.<sup>26,47</sup> To determine the local CompC dose, 1 and 10 mg/kg<sup>48</sup> were tested via local delivery. The 1 mg/kg dose was selected because it inhibited AMPK without altering skin morphology or wound healing (data not shown). For in vitro studies, 1  $\mu$ mol/L CompC was used based on prior studies<sup>49</sup> and validated in the human umbilical vein ECs (HUVEC) scratch wound assay, where it suppressed AMPK activation without affecting wound closure or cell viability (data not shown). The 30 mmol/L glucose concentration ( $\approx$ 500 mg/dL) was selected based on prior EC studies.<sup>50</sup>

## Laser Speckle Contrast Imaging

Blood perfusion at murine wound sites was quantified using laser speckle contrast imaging machine (Perimed) on days 0, 2, and 4 postwounding. Mice were lightly anesthetized with isoflurane, positioned ventral side down and body temperature was maintained at 37°C. Tegaderm dressings were removed before imaging and reapplied afterwards. Wounds were aligned

horizontally for consistent acquisition. All imaging and quantifications were performed using PIMSoft in a blinded manner.

## Histology and Immunostaining

Murine wound tissue and human skin organoids were fixed in formalin for 48 hours, transferred to 70% ethanol, embedded in paraffin, and sectioned. Sections were deparaffinized, rehydrated, and subjected to antigen retrieval in 10 mmol/L Tris (85°C for 30 min). After blocking (5% normal donkey serum, 0.2% Triton-X 100), slides were incubated overnight at 4°C with indicated primary antibodies. Secondary antibodies were incubated at room temperature, and slides were mounted with antifade DAPI. Specificity was confirmed using secondary-only controls (Figure S1). H&E-stained sections were used to measure granulation tissue thickness using ImageJ. Wound borders were defined as 500  $\mu$ m from the edge. Vessel density was quantified as the ratio of Cluster of differentiation 31 (PECAM1 [platelet and endothelial cell adhesion molecule 1]; CD31)+ area to total wound border. Ki-67 proliferation was expressed as the percentage of Ki-67+ nuclei relative to DAPI+ nuclei. Arteriole density was measured as  $\alpha$ -SMA ( $\alpha$ -smooth muscle actin)+ arterioles bigger than 15 to 100  $\mu$ m<sup>51</sup> and normalized to the wound border area. Immunofluorescence and hematoxylin and eosin images were acquired using Olympus FV3000 and Nikon Ti2-E microscope.

## Murine Aortic Ring (Sprout) Assay

Aortas were harvested from 8 to 10 weeks old, male C57BL/6 mice, cleaned from surrounding fat tissue, cut into 1 mm pieces, and embedded into Matrigel in 24-well plates with 0.5 mL growth medium. Growth medium is prepared as described previously.<sup>52</sup> Aortas were then treated with SGLT-2 inhibitors empagliflozin, dapagliflozin, or canagliflozin; 1  $\mu$ mol/L, 1  $\mu$ mol/L, 10  $\mu$ mol/L, respectively or DMSO as vehicle control. Aortic sprouting was imaged using bright-field microscopy for 7 days and harvested at the end point. The total number of branches, number of junctions, and area of sprouting were quantified using ImageJ (National Institutes of Health) and LabKit<sup>53</sup> plug-in.

## Human Skin Organoids

Full-thickness (6 mm) human skin organoids were generated from discarded lower limb amputation tissue from patients with T2D as previously described.<sup>54</sup> Circular 3-mm full-thickness wounds were created and organoids embedded in collagen I and maintained at an air–liquid interface in DMEM (Dulbecco's Modified Eagle Medium) supplemented with 10 mmol/L HEPES (4-[2-hydroxyethyl]-1-piperazineethanesulfonic acid), 50  $\mu$ g/mL ascorbic acid, 100  $\mu$ mol/L adenine, 0.5  $\mu$ mol/L hydrocortisone, 0.1 nmol/L cholera toxin, 100 U/mL penicillin/streptomycin with high glucose (30 mmol/L) or osmotic control (5 mmol/L glucose + 25 mmol/L mannitol). Organoids were treated every 2 days with empagliflozin, dapagliflozin, canagliflozin (1  $\mu$ mol/L, 1  $\mu$ mol/L, and 10  $\mu$ mol/L, respectively), or vehicle (DMSO). Wound closure was imaged and analyzed daily for 5 days by Live-Cell Imager (Axion). Samples were obtained from 3 male patients, ages 48 to 65, with T2D.

## Human Microvessel Sprout Assay

Human subcutaneous vessels with an approximate diameter of  $\approx$ 500  $\mu$ m were isolated from discarded nonulcerated skin

obtained from T2D amputations (n=3 male donors with T2D ages 48–65). Small vessels trimmed and cut into 1 mm segments, embedded individually in Matrigel. Growth media<sup>52</sup> with high glucose (30 mmol/L) were changed every 2 days and microvessels were imaged every 2 days to visualize sprouting with bright field microscopy. Microvessels with sprouts were harvested 10 days later. The sprouting area, number of branches, and number of junctions were quantified using ImageJ with LabKit<sup>53</sup> plug-in.

## Cell Culture and In Vitro Experiments

HUVECs passaged  $<5\times$  were used for all experiments and cultured in EGM-2 media. For SGLT2 inhibitor treatment studies, HUVECs were treated with empagliflozin, dapagliflozin, canagliflozin and CompC prepared as final concentrations in either normal glucose (5 mmol/L) with addition of mannitol (25 mmol/L) or 30 mmol/L of high glucose conditioned media, unless specified otherwise as 1  $\mu$ mol/L, 1  $\mu$ mol/L, 10  $\mu$ mol/L, and 1  $\mu$ mol/L, respectively. The final concentration of DMSO was 0.1% for in vitro experiments.

## Scratch Assay for EC Migration

HUVECs were plated in 12-well plates at 50 000 cells/well and treated with SGLT2 inhibitors empagliflozin, dapagliflozin or canagliflozin; at final concentrations of 1  $\mu$ mol/L, 1  $\mu$ mol/L, 10  $\mu$ mol/L and with CompC at 1  $\mu$ mol/L, respectively, and cultured for 48 hours under mannitol with normal glucose or high glucose conditions. The cell monolayer was scratched using a 200  $\mu$ L micropipette tip to form a 750  $\mu$ m wound, and the wound was imaged by the live-cell imager for 24 hours to assess wound closure.

## EdU Proliferation Assay

HUVECs were plated in 24-well plates (12 000 cells/well) and treated with empagliflozin, dapagliflozin, or canagliflozin (1, 1, and 10  $\mu$ mol/L, respectively) under high-glucose conditions for 48 hours. Cells were labeled with EdU using Click-iT and incubated for 18 hours. After fixation (4% paraformaldehyde), permeabilization (0.5% Triton X-100), and Click-iT reaction, nuclei were counterstained with Hoechst 33342. Images were acquired by Nikon fluorescent microscope, and proliferation was quantified as the percentage of EdU-positive nuclei relative to total Hoechst-positive nuclei using ImageJ.

## HUVEC Permeability Assay

HUVECs were seeded at  $1\times 10^5$  cells per 0.4- $\mu$ m Transwell insert and cultured for 72 hours to form confluent monolayers. Cells were exposed to high glucose (30 mmol/L) or osmotic control (5 mmol/L glucose + 25 mmol/L mannitol) and treated with empagliflozin, dapagliflozin, canagliflozin, or vehicle for 48 hours. FITC (fluorescein isothiocyanate)-dextran (0.1 mg/mL) was added to the upper chamber, and fluorescence in the lower chamber was measured after 60 minutes (Ex 494 nm/Em 518 nm). Concentrations were calculated from a standard curve.

## Vascular Permeability Analysis

Six days postwounding, 200  $\mu$ L of FITC/dextran (15 mg/mL in PBS) was administered via retro-orbital injection to db/db mice 30 minutes before euthanasia.<sup>55</sup> Wound tissues were

embedded in OCT (optimal cutting temperature compound), cryosectioned (10  $\mu$ m), and mounted with DAPI-containing anti-fade medium.<sup>56</sup> FITC-dextran extravasation was quantified by thresholding fluorescence images acquired under identical exposure settings and calculating FITC-positive area relative to the total wound area using ImageJ. Background fluorescence was uniformly subtracted before analysis.

## RNA Extraction and Real-Time Quantitative Polymerase Chain Reaction

Snap-frozen mouse and human tissues were homogenized in TRIzol, and total RNA was isolated according to the manufacturer's instructions. RNA was separated using chloroform, precipitated with isopropanol, and resuspended in RNA-free water. Reverse transcription and quantitative polymerase chain reaction were performed. Gene-specific primers are listed in Table S1. Samples were normalized to  $\beta$ -Actin. Fold changes were calculated using  $\Delta\Delta$ Ct method.

## Western Blot Analysis

Mouse tissues were homogenized and lysed in RIPA (radioimmunoprecipitation assay) buffer supplemented with protease inhibitors and EDTA. Lysates were clarified by centrifugation at 15 000 rpm for 15 minutes. Protein quantification was performed by BCA assay. Lysates were separated using 4% to 15% TGX mini-PROTEAN Gels, transferred to PVDF (polyvinylidene difluoride) membranes, and incubated with indicated primary antibodies (1:1000 dilution) followed by HRP-conjugated secondary antibodies (1:2000 dilution). ECL assay was performed per manufacturer's instructions. Western blotting for phospho-AMPK and AMPK was performed using antibodies validated for specificity with positive and negative control cell extracts. All blots represent  $n=5$  to 7 biological replicates from  $\geq 3$  independent experiments for in vitro or from  $n=6$  mice/group, 1 to 2 wounds/mouse for in vivo studies. Controls (DMSO as vehicle, and mannitol for osmotic control) were included and blots were imaged to avoid saturation. Quantification was performed using ImageJ.

## Quantification and Statistical Analysis

Data are expressed as the mean  $\pm$  SEM. Normality and homoscedasticity were assessed using the Shapiro-Wilk and Bartlett test respectively. Parametric data were analyzed by 1-way or 2-way ANOVA with Tukey posthoc test for multiple comparisons. Nonparametric data were analyzed using the Kruskal-Wallis test. Statistical significance was set at a  $P < 0.05$ . All analyses included data from  $\geq 5$  independent biological replicates across in vivo, ex vivo, and in vitro unless otherwise noted and were performed using GraphPad PRISM.

# RESULTS

## Empagliflozin and Dapagliflozin, But Not Canagliflozin, Improved Wound Healing, Perfusion, and Granulation Tissue Formation In Vivo

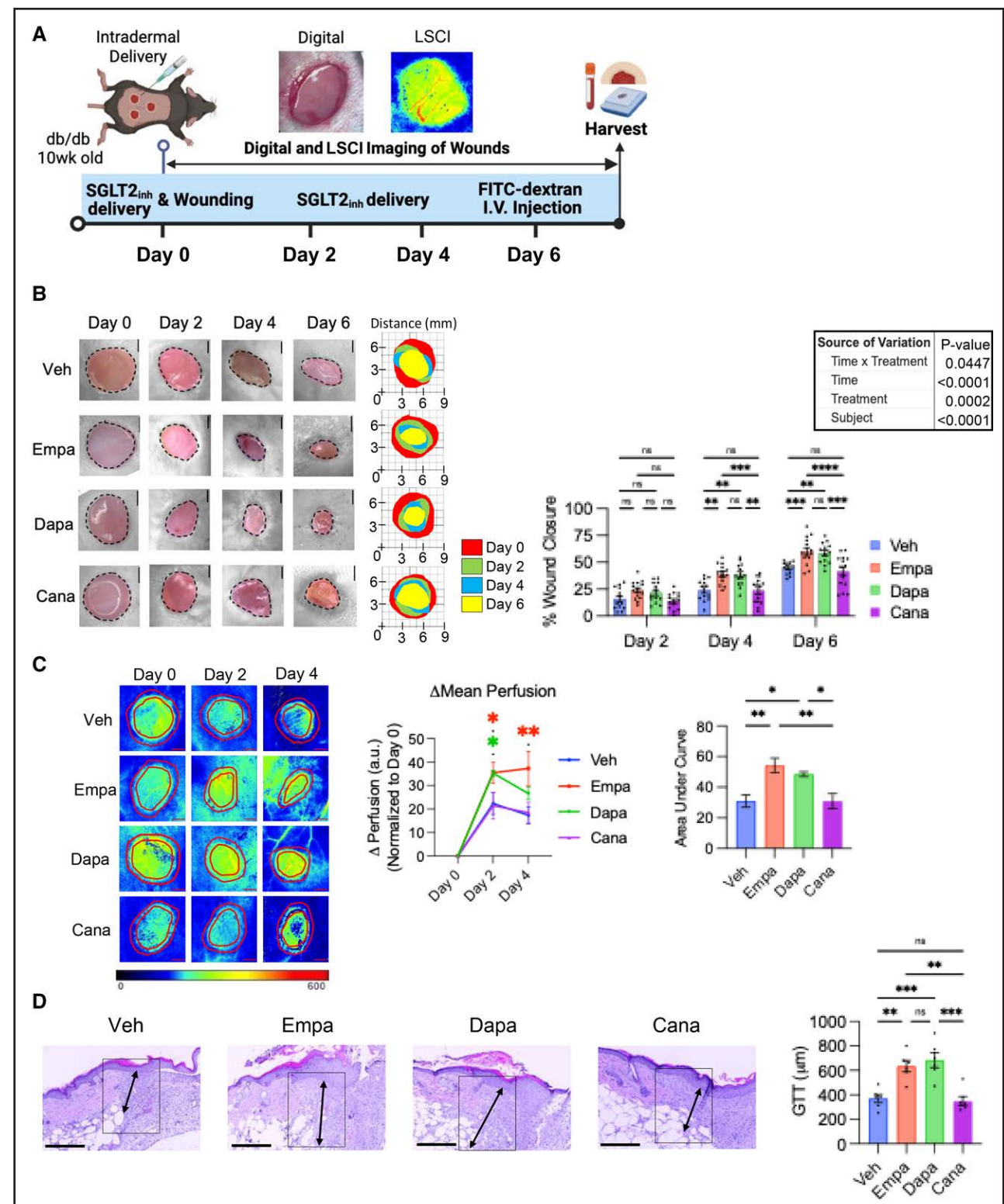
To assess the effects of individual SGLT2 inhibitors on diabetic cutaneous wound healing, wounds were

generated on the dorsal skin of db/db mice. Empagliflozin, dapagliflozin, canagliflozin, or a vehicle control was delivered intradermally at the wound edges (Figure 1A). No significant changes in blood glucose, body weight, or total plasma free fatty acids levels were observed during the study (Figure S2). Wound closure increased by 31% and 30% in the empagliflozin- and dapagliflozin- treated mice, respectively, while canagliflozin showed no improvement compared with vehicle at day 6 (Figure 1B). Laser speckle contrast imaging demonstrated increased blood flow at the wound margins in empagliflozin- and dapagliflozin- treated mice, with perfusion increased by 58% and 56% on day 2 and by 115% and 54% on day 4, respectively, relative to day 0; no significant increase was observed in canagliflozin-treated mice. (Figure 1C). Histological analysis showed increased granulation tissue thickness with empagliflozin and dapagliflozin (75% and 84%, respectively), with no significant effect with canagliflozin compared with vehicle controls (Figure 1D). To determine whether these effects were specific to diabetic conditions, we evaluated local SGLT2 inhibitor delivery in normoglycemic littermate controls (Het db/+). In this setting, none of the SGLT2 inhibitors improved wound closure or perfusion compared with vehicle, and no significant differences were observed among treatment groups (Figure S3). These data indicate that the prohealing effects of empagliflozin and dapagliflozin are context-dependent and are amplified under diabetic conditions.

## Empagliflozin and Dapagliflozin Increase Angiogenesis and Endothelial Proliferation In Vivo and in Human Skin Organotypic Cultures

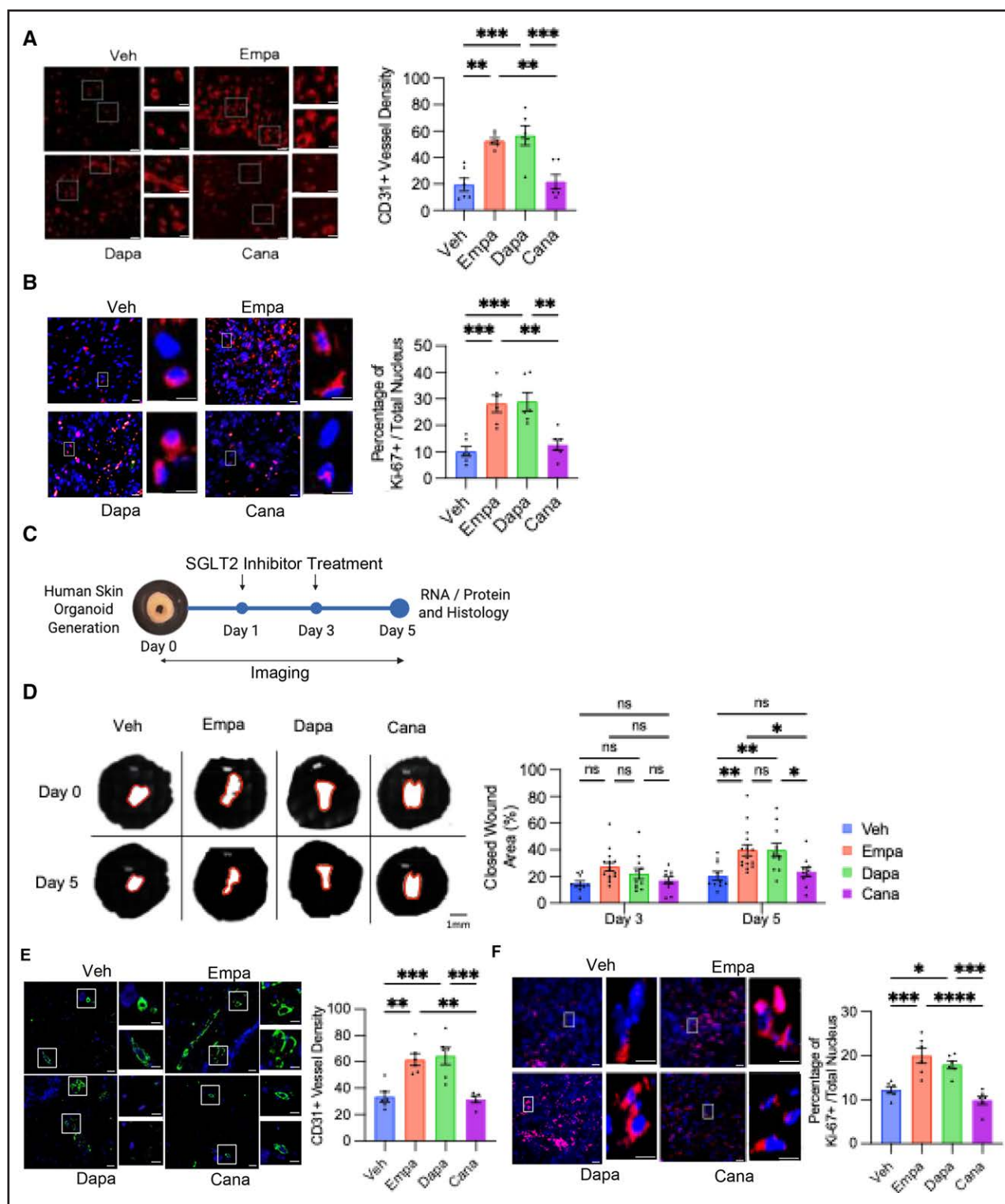
To determine whether SGLT2 inhibitors promote neovascularization during wound healing, we assessed vascular density and cellular proliferation at the wound border. Consistent with improvement in wound closure, CD31<sup>+</sup> (PECAM1<sup>+</sup>) vessel density increased 2.08- and 2.26-fold in response to empagliflozin and dapagliflozin, respectively, without a significant change in response to canagliflozin at day 6 (Figure 2A). Ki-67 immunostaining confirmed increased proliferation at the wound border with empagliflozin and dapagliflozin (1.62- and 1.73-fold, respectively) compared with vehicle, with no effect of canagliflozin (Figure 2B). Arteriolar density, defined by  $\alpha$ -SMA staining of vessels 15 to 100  $\mu$ m in diameter, did not differ among treatment groups, indicating that SGLT2 inhibitors do not alter arteriole number during the early phase of wound healing (Figure S4).

To validate these findings in human tissue, we used an ex vivo human skin organotypic wound model generated from discarded tissue samples from patients with T2D (Figure 2C), as previously described.<sup>54</sup> Empagliflozin and



**Figure 1. Empagliflozin (EMPA) and dapagliflozin (DAPA), but not canagliflozin (CANA), improve wound closure, perfusion, and granulation tissue formation in diabetic mice.**

Db/db mice were treated with vehicle (VEH), EMPA, DAPA, or CANA by intradermal delivery. **A**, Schematic of experimental design. **B**, Representative images and quantification of wound closure at days 2, 4, and 6 postinjury. **C**, Wound border perfusion assessed by laser speckle imaging (scale bar: 2 mm; N=6 mice, 3 wounds/mouse). **D**, Granulation tissue thickness (GTT) measured in H&E-stained wound sections at day 6 (scale bar: 300 μm; N=6 mice/group). Statistical significance was determined by 1-way or 2-way ANOVA with Tukey post hoc analysis. \* $P<0.05$ , \*\* $P<0.01$ , \*\*\* $P<0.001$ , \*\*\*\* $P<0.0001$ . Data are presented as mean±SEM. LSCI indicates laser speckle contrast imaging; and SGLT<sub>2</sub>, sodium-glucose cotransporter 2. Figure 1A created in BioRender. Bestepe, F. (2026) <https://BioRender.com/ztr5zwt>.



**Figure 2. Empagliflozin (EMPA) and dapagliflozin (DAPA), but not canagliflozin (CANA), enhance angiogenesis and proliferation in db/db mice wounds and human skin organoids.**

**A** and **B**, Db/db mice were treated with intradermal vehicle (VEH), EMPA, DAPA, or CANA. **A**, Microvessel density in mouse wound tissue was measured by CD31 confocal immunofluorescence. **B**, Cell proliferation in granulation tissue was quantified by Ki-67+ nuclei using confocal immunofluorescence (N=6 mice/group). **C**, Schematic depicting generation of human organotypic skin wounds. **D**, Representative images of wound closure 5 days after wounding (n=10–12 organoids/group). **E** and **F**, Angiogenesis and proliferation in ex vivo wounds were measured by CD31 and Ki-67 confocal immunofluorescence, respectively (n=6 organoids/group). Scale bars: 25  $\mu$ m, Inset scale bars: 5  $\mu$ m. Statistical significance was determined by 1-way ANOVA with Tukey post hoc analysis.  $P < 0.05$ ,  $P < 0.01$ ,  $P < 0.001$ ,  $P < 0.0001$ . Data are presented as mean  $\pm$  SEM. SGLT2 indicates sodium-glucose cotransporter 2. Figure 2C created in BioRender. Bestepe, F. (2026) <https://BioRender.com/ztr5zwt>.

dapagliflozin significantly improved wound closure (79% and 95%, respectively) by day 5, whereas canagliflozin showed no effect compared with vehicle (Figure 2D). Consistent with these results, empagliflozin and dapagliflozin increased angiogenesis as measured by CD31 expression, and cell proliferation as measured by Ki67<sup>+</sup> nuclei, while canagliflozin had no significant effect (Figure 2E and 2F).

To evaluate dose sensitivity, we conducted dose-response experiments in human skin organoids under high glucose conditions. High glucose (30 mmol/L) impaired wound closure, whereas the osmotic control (25 mmol/L mannitol + 5 mmol/L glucose) had no effect. Increasing empagliflozin and dapagliflozin beyond 1  $\mu$ M conferred no additional benefit, while reducing the dose to 0.5  $\mu$ M attenuated efficacy relative to the 1  $\mu$ M dose. In contrast, canagliflozin exhibited a distinct dose response pattern, with only at 2  $\mu$ M and no improvement at 20  $\mu$ M (Figure S5). These findings underscore the distinct dose-sensitivity profiles and therapeutic windows among individual SGLT2 inhibitors in diabetic wound healing.

### SGLT2 Inhibitors Modulate Inflammation and Fibroblast Activation, With Empagliflozin and Dapagliflozin Showing Enhanced Reparative Signatures

Expression of proinflammatory cytokine and enzyme genes, including *TNF* (tumor necrosis factor- $\alpha$ ), *IL1B* (interleukin-1 $\beta$ ), *IL6* (interleukin-6), *CCL2* (MCP-1 [monocyte chemoattractant protein-1]), and *PTGS2* (COX-2 [cyclooxygenase-2]), was significantly decreased in all 3 SGLT2 inhibitor-treated groups compared with vehicle in both in vivo and ex vivo systems (Figure 3A and 3C). In parallel, expression of EC adhesion molecule genes, including *ICAM1* (intercellular adhesion molecule 1), *VCAM1* (vascular cell adhesion molecule 1), and *SELE* (E-selectin), was also significantly decreased at the wound borders of db/db mice and in human skin organoids (Figure 3B and 3D), indicating a shared anti-inflammatory effect. However, assessment of fibroblast-associated gene expression revealed differential regulation among the SGLT2 inhibitors. Empagliflozin and dapagliflozin significantly increased key fibroblast-associated genes involved in granulation and tissue remodeling, including *COL1A1* (collagen type I alpha 1 chain), *ACTA2* ( $\alpha$ -SMA), *VIM* (vimentin), *MMP9* (matrix metalloproteinase 9), *TGFB1* (TGF- $\beta$  [transforming growth factor- $\beta$ ]), and *ITGA5* (integrin subunit alpha 5) mRNA expression, whereas canagliflozin did not significantly increase these markers. Changes in *MMP2* and *COL3A1* expression did not differ significantly among treatment groups (Figure 3E). Together, these data suggest that while all SGLT2 inhibitors suppress inflammatory and EC activation markers, only empagliflozin and

dapagliflozin significantly increased expression of genes associated with fibroblast activation and tissue remodeling, consistent with a reparative phenotype.

### Empagliflozin and Dapagliflozin Enhance, But Canagliflozin Impairs, Sprouting Angiogenesis, Endothelial Migration and Proliferation, Whereas All SGLT2 Inhibitors Improve Barrier Integrity

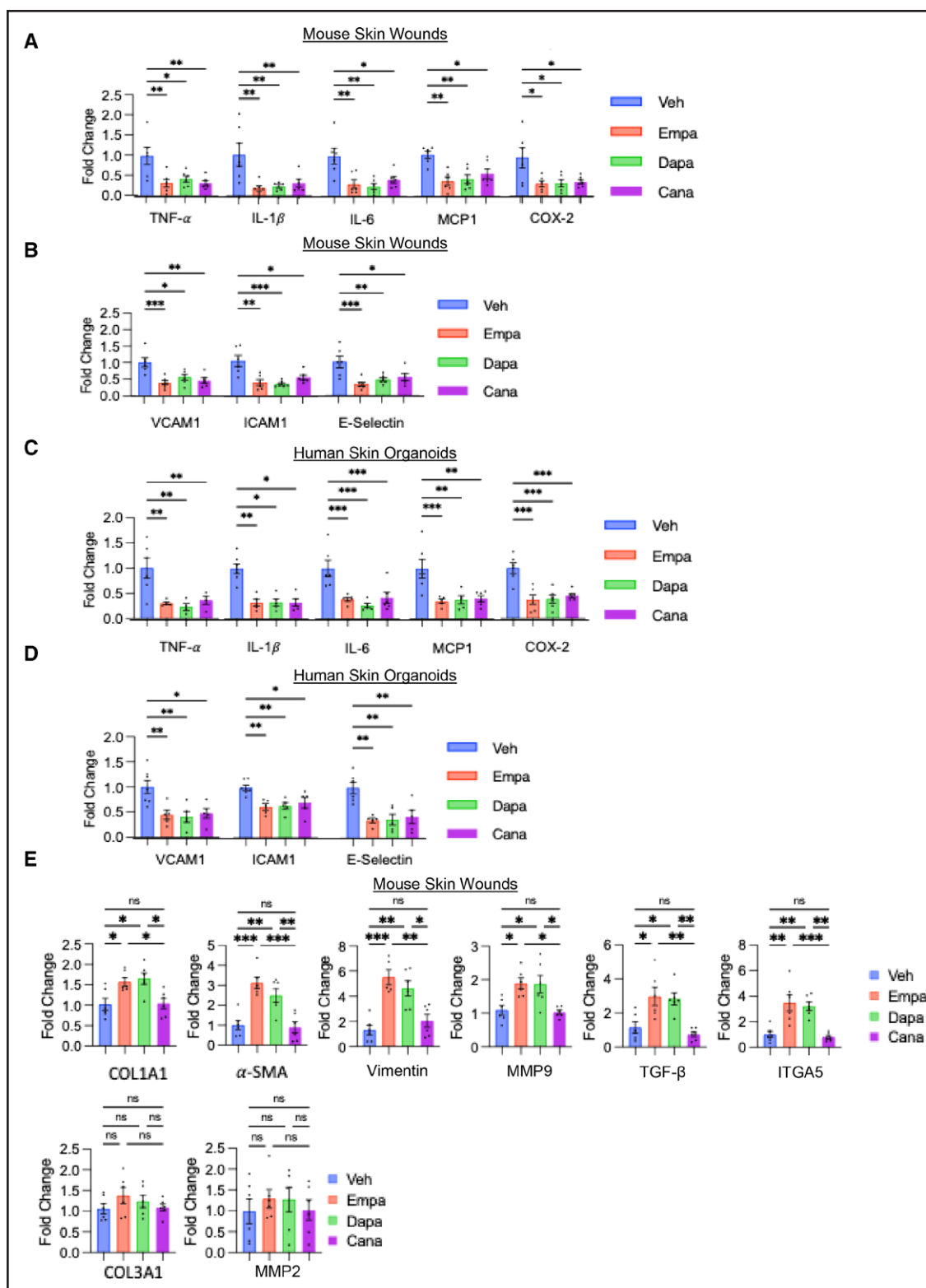
To examine sprouting angiogenesis ex vivo, subcutaneous microvessel explants from patients with T2D were embedded in Matrigel and treated with empagliflozin, dapagliflozin, or canagliflozin. Empagliflozin and dapagliflozin significantly increased branch formation, junctions, and total sprout area compared with vehicle, whereas canagliflozin markedly reduced junction formation without affecting branch number or total sprout area (Figure 4A).

Similarly, murine aortic rings under high-glucose conditions showed enhanced sprouting with empagliflozin and dapagliflozin, reflected by increased branch formation, vascular junctions, and total sprout area compared with vehicle, while canagliflozin reduced branch and junction formation (Figure 4B). These findings indicate that empagliflozin and dapagliflozin promote, but canagliflozin impairs sprouting angiogenesis.

To examine the effects of SGLT2 inhibitors on additional EC-angiogenic functions, we performed in vitro scratch wound healing assays with EdU labeling of HUVEC monolayers treated with empagliflozin, dapagliflozin, canagliflozin, or vehicle under high glucose for 48 hours. Empagliflozin and dapagliflozin significantly increased EC migration by 15.3% and 22.5% and proliferation by 39.6% and 29.8%, respectively. In contrast, canagliflozin impaired both responses, reducing migration and proliferation, by 21% and 27%, respectively (Figure 4C and 4D). To exclude osmotic stress as a confounding factor, we used 25 mmol/L mannitol as an osmotic control, confirming that the inhibition of EC wound closure was attributable to high glucose (Figure S6).

Additionally, empagliflozin and dapagliflozin significantly increased endothelial gene markers associated with vascular homeostasis and angiogenesis, including *Pecam1*, *Vwf* (von Willebrand factor), *Cdh5*, *Angpt1* (angiopoietin 1), *Kdr* (kinase insert domain receptor), *Nos3*, and *Tie2* (Tek), compared with vehicle, while downregulating the antiangiogenic factor *Thbs1* (thrombospondin 1). *Angpt2* (angiopoietin 2) expression was not significantly altered by any SGLT2 inhibitor. In contrast, canagliflozin had minimal effect on most markers and failed to suppress *Thbs1* expression (Figure 4E).

We next examined the effects of SGLT2 inhibitors on endothelial barrier function, a key vascular complication in diabetes.<sup>57</sup> Under high-glucose conditions, HUVECs exhibited a 1.54-fold increase in FITC-dextran flux compared



**Figure 3. SGLT2 (sodium-glucose cotransporter 2) inhibitors reduce inflammation and endothelial activation, while empagliflozin (EMPA) and dapagliflozin (DAPA) selectively enhance wound repair gene expression.**

**A**, Indicated proinflammatory cytokines and **(B)** endothelial adhesion molecules in mouse wounds ( $n=6$  mice/group); and **C** and **D** in human skin organoids ( $n=5-7$  organoids/group) were measured by reverse transcription quantitative polymerase chain reaction (RT-qPCR). **E**, Expression of genes associated with fibroblast activation, extracellular matrix remodeling, and wound repair were measured in skin wounds by RT-qPCR ( $n=6$  mice/group). Statistical significance was determined by 1-way ANOVA with Tukey post hoc analysis. \* $P<0.05$ , \*\* $P<0.01$ , \*\*\* $P<0.001$  per condition. Data are presented as mean $\pm$ SEM. ACTA2 indicates actin alpha 2, smooth muscle; CCL2, C-C motif chemokine ligand 2 (also known as MCP-1); COL1A1, collagen type I alpha 1 chain; COL3A1, collagen type III (Continued)

with normal-glucose controls, consistent with impaired barrier integrity. Treatment with empagliflozin, dapagliflozin, or canagliflozin significantly restored barrier function, reducing FITC-dextran leakage by 33%, 30.1%, and 18.3%, respectively, compared with vehicle (Figure 4F). To extend these findings in vivo, vascular permeability was assessed at wound borders; empagliflozin, dapagliflozin and canagliflozin reduced FITC-dextran extravasation by 57%, 48%, and 42%, respectively, indicating improved barrier integrity across all 3 inhibitors (Figure 4G).

### Canagliflozin-Mediated Differential Regulation of AMPK/AKT/mTORC1 Signaling Impairs EC Function

AMPK exerts dual effects on EC angiogenesis, supporting vascular homeostasis at moderate activation<sup>58</sup> but suppressing growth when overactivated.<sup>32,59</sup> In murine skin wound tissue (day 6), empagliflozin and dapagliflozin induced moderate AMPK phosphorylation (to 1.81- and 1.73-fold), while preserving AKT signaling and mTORC1 downstream targets, RPS6 (ribosomal protein S6) and eukaryotic translation initiation factor 4E-BP1 (4E-binding protein 1) compared with vehicle. In contrast, canagliflozin caused marked AMPK activation (4.1-fold), accompanied by suppression of AKT, RPS6 and 4E-BP1 phosphorylation (73%, 55%, and 85%) relative to empagliflozin and dapagliflozin, and a mild decrease (48%, 23%, and 51%) compared with vehicle (Figure 5A).

Similarly, under high-glucose conditions, HUVECs treated with empagliflozin and dapagliflozin exhibited moderate AMPK phosphorylation (1.81- and 1.73-fold), while maintaining AKT, RPS6, and 4E-BP1 phosphorylation. In contrast, canagliflozin further increased AMPK phosphorylation by 4.28-fold while suppressing AKT, RPS6 and 4E-BP1 phosphorylation (34%, 52%, and 36%, respectively) compared with vehicle (Figure 5B). AMPK-positive and -negative control cell lysates confirmed antibody specificity, mannitol controls confirmed that AMPK suppression was glucose-dependent rather than osmotic, and all 3 SGLT2 inhibitors restored AMPK phosphorylation, with canagliflozin showing the greatest increase (Figure S7). AMPK activity was further assessed by phosphorylation of ACC (acetyl-CoA carboxylase), a canonical AMPK target.<sup>60</sup> ACC phosphorylation increased in wound tissue (1.46-, 1.75-, and 4.34-fold) and HUVECs (1.78-, 1.96-, and 3.43-fold) following empagliflozin, dapagliflozin and canagliflozin, respectively (Figure 5A, wound tissue; Figure 5B, HUVECs). Broader stress-responsive cascades (p38 MAPK [p38 mitogen-activated protein kinase], ERK1/2 [extracellular

signal-regulated kinase 1/2], and JNK [c-Jun N-terminal kinase]) were similarly induced by glucose but equally attenuated by all 3 SGLT2 inhibitors without significant differences among agents (Figure S8).

To examine whether excessive AMPK activation impairs endothelial angiogenic function, we treated HUVECs with canagliflozin and coadministered CompC, an AMPK inhibitor. Canagliflozin (10  $\mu$ mol/L) impaired EC scratch wound closure by 36.5%, and CompC partially rescued this phenotype, improving wound closure by 21.6% compared with vehicle (Figure 5C). Canagliflozin increased AMPK phosphorylation by 2.41-fold while suppressing downstream AKT, RPS6 and 4E-BP1 phosphorylation. CompC (1  $\mu$ mol/L) attenuated AMPK phosphorylation by 78% while restoring AKT, RPS6, and 4E-BP1 phosphorylation compared with canagliflozin (Figure 5D). These findings indicate that sustained AMPK activation with concomitant suppression of AKT/mTORC1 signaling paradoxically impairs EC angiogenic function under canagliflozin treatment.

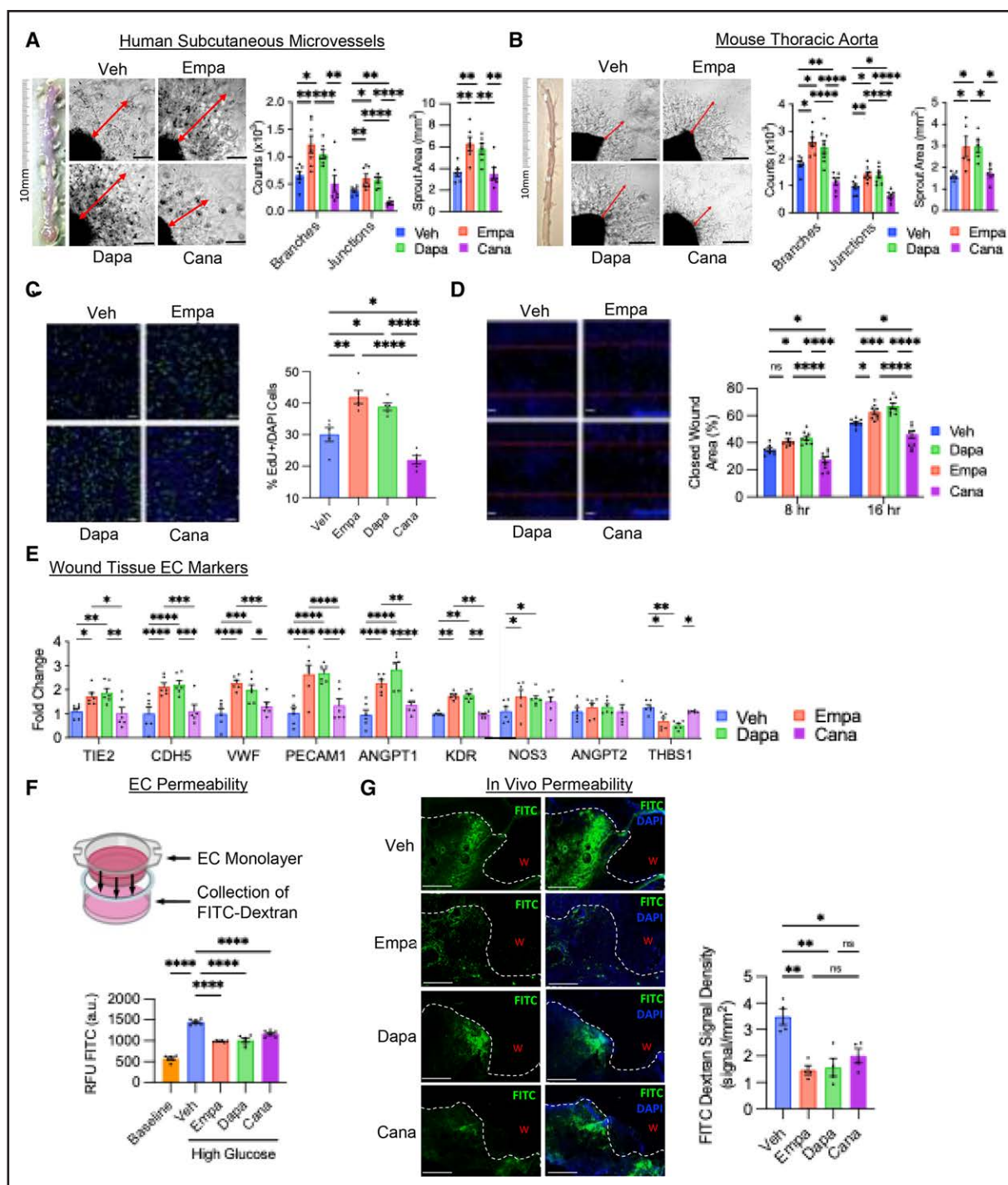


### Lower Canagliflozin Dosing and Pharmacological AMPK Inhibition Improved Canagliflozin's Role in Wound Healing and Restored Angiogenic Signaling In Vivo

To determine whether canagliflozin's limited efficacy reflects excessive AMPK activation, we reduced canagliflozin dosing or cotargeted AMPK signaling by CompC (Figure 6A). Reducing canagliflozin to 1 mg/kg (equivalent to empagliflozin/dapagliflozin dosing) partially improved wound closure by 38% at day 6, increased perfusion by 42%, and improved granulation tissue thickness by 86% relative to canagliflozin 10 mg/kg (Figure 6A through 6C). Codelivery of CompC with 10 mg/kg canagliflozin significantly improved wound closure by 65% at day 4 and 40% at day 6, increased perfusion by 75%, and improved granulation tissue thickness by 61% compared with canagliflozin alone (Figure 6A through 6C). Both interventions increased expression of angiogenesis-related EC markers, including *Vwf*, *Pecam1*, *Angpt1*, *Angpt2*, and *Kdr*, with a modest increase in *Tek* (also known as *Tie2*) compared with canagliflozin 10 mg/kg. Codelivery with CompC reduced expression of anti-angiogenic marker gene *Thbs1* (Figure 6D).

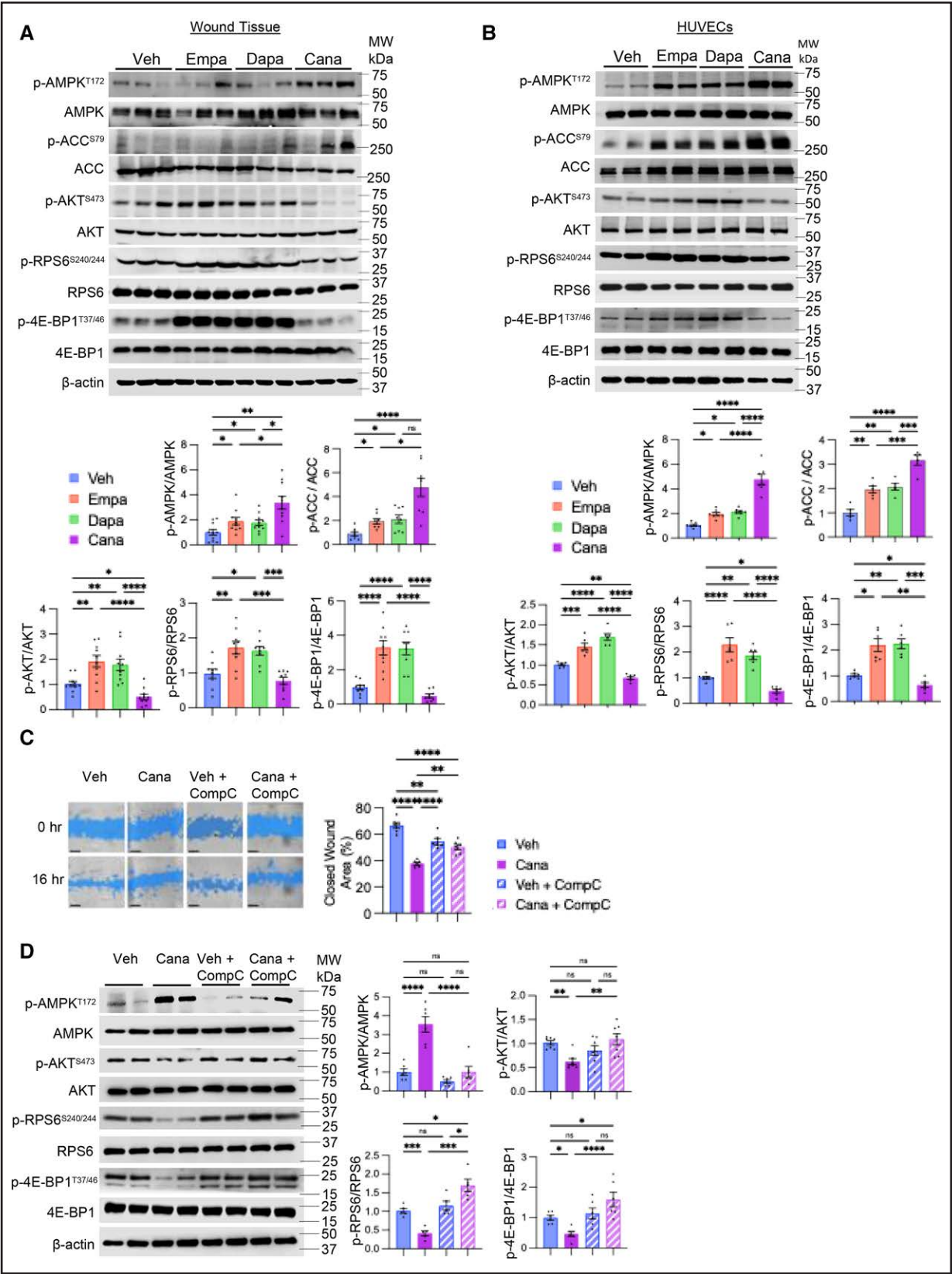
Mechanistically, both low-dose canagliflozin and codelivery of canagliflozin 10 mg/kg dose with 1 mg/kg CompC (canagliflozin+ CompC) decreased excessive AMPK phosphorylation by 64% and 86%, respectively, compared with 10 mg/kg canagliflozin alone. As

**Figure 3 Continued.** *alpha 1 chain*; *ICAM1*, intercellular adhesion molecule 1; *IL1B*, interleukin 1 beta; *ITGA5*, integrin subunit alpha 5; *MMP2*, matrix metalloproteinase 2; *MMP9*, matrix metalloproteinase 9; *PTGS2*, prostaglandin-endoperoxide synthase 2 (also known as COX-2); *TGFB1*, transforming growth factor beta 1; *TNF*, tumor necrosis factor; *VCAM1*, vascular cell adhesion molecule 1; and *VIM*, vimentin. Figure 3F created in BioRender. Bestepe, F. (2026) <https://BioRender.com/ztr5zw7>.



**Figure 4. Empagliflozin (EMPA) and dapagliflozin (DAPA) improve endothelial angiogenic programs and endothelial cell (EC) function, while canagliflozin (CANA) shows limited benefit.**

**A** and **B**, Vessel sprouting was quantified as absolute counts of branches and junctions, and as area (mm<sup>2</sup>) in human subcutaneous microvessels (**A**, day 7) and mouse thoracic aorta (**B**, day 5) treated with vehicle (VEH), EMPA, DAPA, or CANA (n=5–8 independent biological samples; scale bars: 500  $\mu$ m). **C** and **D**, Human umbilical vein endothelial cells (HUVECs) treated with VEH, EMPA, DAPA, or CANA were subjected to (**C**) EdU proliferation and (**D**) scratch-wound migration assays (n=5–8 biological replicates from 2 independent experiments; scale bars: 100  $\mu$ m). **E**, Angiogenesis-related genes in wound samples was quantified by reverse transcription polymerase chain reaction. **F**, FITC (fluorescein isothiocyanate)-dextran permeability of HUVEC monolayers under high glucose (n=5 technical replicates from 3 independent experiments). **G**, FITC-dextran extravasation in mouse wounds 30 mins after i.v. delivery, (W: wound area; n=4 mice from n=2 independent experiments, scale bars: 300  $\mu$ m). Statistical significance was determined by 1-way ANOVA with Tukey post hoc analysis. \* $P$ <0.05, \*\* $P$ <0.01, \*\*\* $P$ <0.001, \*\*\*\* $P$ <0.0001. Data are presented as mean $\pm$ SEM. *ANGPT1* indicates angiotensinogen 1; *CDH5*, cadherin 5; FITC, fluorescein isothiocyanate; *KDR*, kinase insert domain receptor; *NOS3*, nitric oxide synthase 3; *PECAM1*, platelet and endothelial cell adhesion molecule 1; *THBS1*, thrombospondin 1; *TIE2/TEK*, TEK receptor tyrosine kinase (also known as TIE2); and *VWF*, von Willebrand factor.



**Figure 5. SGLT2 (sodium-glucose cotransporter 2) inhibitors differentially regulate AMPK (AMP-activated protein kinase)/AKT (AKT serine/threonine kinase)/mTORC1 (mechanistic target of rapamycin complex 1) signaling in endothelial cells (ECs) in an AMPK-dependent manner.**  
**A**, Phosphorylation of AMPK, ACC (acetyl-CoA carboxylase), AKT, RPS6 (ribosomal protein S6), and 4E-BP1 (eukaryotic translation initiation factor 4E-binding protein 1) was quantified in db/db mouse wounds by western blot. (N=6 mice, 1–2 wounds/group). (Continued)

AMPK activation decreased, AKT/mTORC1 signaling was restored. Notably, low-dose canagliflozin and canagliflozin+CompC increased phosphorylation of AKT, RPS6, and 4E-BP1, consistent with restoration of AKT/mTORC1 signaling (Figure 6E). Interestingly, low-dose canagliflozin led to a modest 26% increase in AMPK phosphorylation, whereas CompC cotreatment reduced AMPK phosphorylation to  $\approx 50\%$  below vehicle levels, indicating a more substantial effect on restoring pathway balance. These results underscore AMPK overactivation as a key driver of canagliflozin's impaired ability to promote endothelial repair and angiogenesis. By lowering its dose or pharmacologically inhibiting AMPK, canagliflozin's proangiogenic and prohealing functions were significantly restored, revealing a previously unrecognized in-class difference among SGLT2 inhibitors and highlighting AMPK overactivation as a key mechanistic driver of canagliflozin's limited efficacy.

Taken together, empagliflozin and dapagliflozin consistently enhanced wound healing and angiogenesis in vitro, ex vivo, and in vivo models, whereas canagliflozin showed limited reparative benefit and impaired angiogenic responses. These divergent outcomes were associated with differential regulation of AMPK/AKT/mTORC1 signaling, suggesting that the balance, rather than activation alone, of these pathways determines vascular repair capacity in the diabetic wound healing microenvironment.

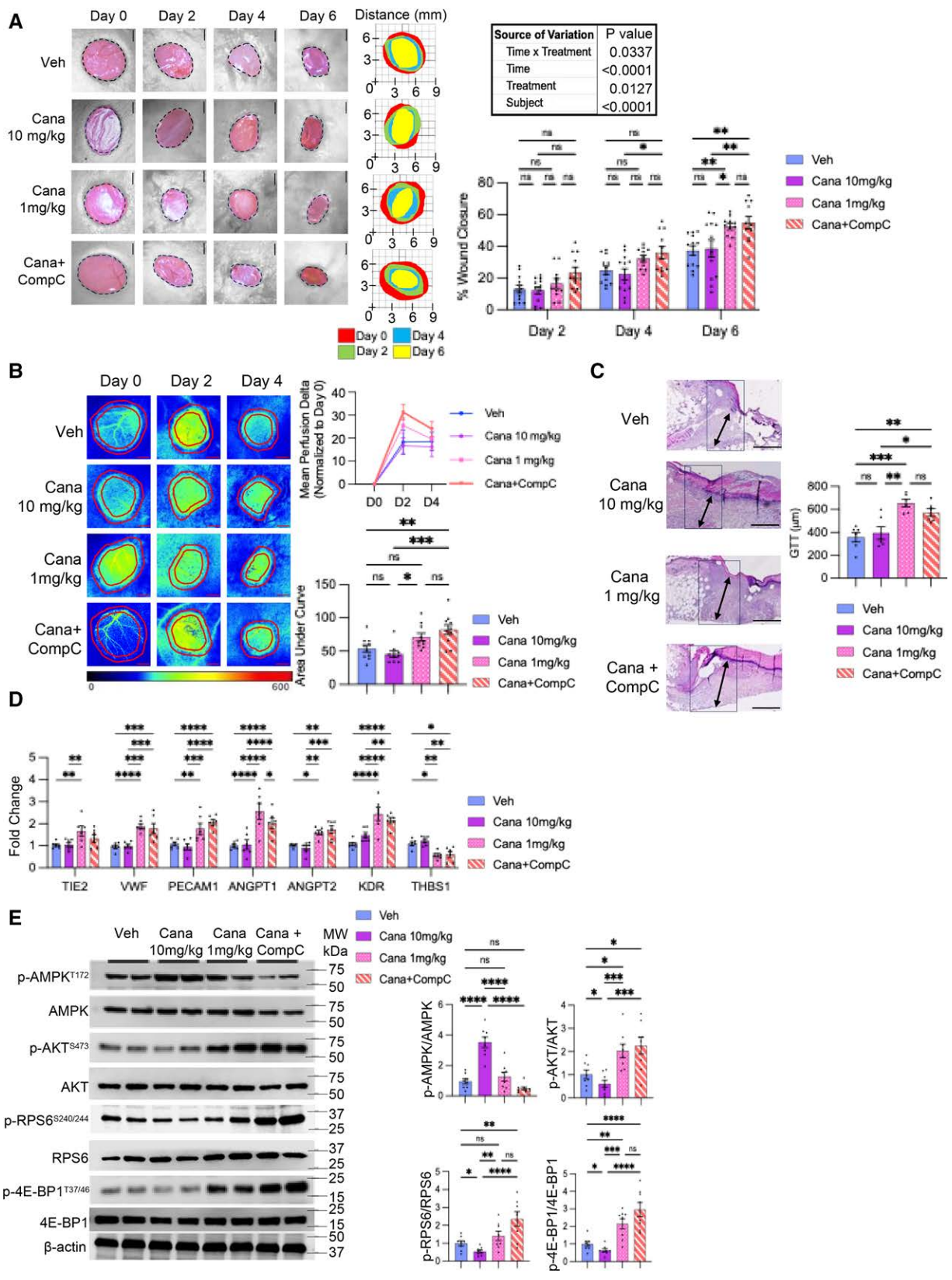
## DISCUSSION

Our findings highlight key differences in how individual SGLT2 inhibitors regulate endothelial repair and angiogenesis under diabetic conditions: (1) empagliflozin and dapagliflozin significantly improved wound healing, perfusion, and angiogenesis in db/db mice; (2) both agents enhanced EC function in ex vivo human and murine sprouting assays reflected by increasing branch and junction formation; (3) empagliflozin and dapagliflozin promoted EC proliferation and migration under hyperglycemic conditions in vitro; (4) canagliflozin did not improve angiogenesis or wound closure in diabetic murine wounds or human ex vivo organoid models; (5) canagliflozin impaired microvascular sprouting, EC proliferation, and migration under hyperglycemic conditions; and (6) these effects were associated with decreased AKT and mTORC1 signaling despite increased AMPK activation. Together, empagliflozin and dapagliflozin

may support vascular regeneration through coordinated AMPK activation while maintaining AKT-mTORC1 signaling, whereas canagliflozin disrupts this balance, potentially impairing EC function. By integrating murine and human organoid-based models, we provide robust cross-species validation, strengthening the translational relevance of SGLT2 inhibitor-specific local vascular effects. Our findings suggest that pro-angiogenic effects observed with empagliflozin and dapagliflozin may be partially mediated by upregulation of the Angpt1/Tie2 signaling axis, including a greater induction of Angpt1 (Figure 4E), which promotes angiogenesis and stabilizes capillary networks. Accordingly, empagliflozin and dapagliflozin increased Angpt1 and Tie2 gene expression in diabetic wounds, consistent with enhanced angiogenesis and barrier function (Figures 2A and 4G). Given the reported crosstalk between AMPK/AKT signaling and Angpt1/Tie2 activation,<sup>61,62</sup> these pathways may converge to support EC function. In contrast, canagliflozin did not induce Angpt1/Tie2 mRNA expression, consistent with its blunted angiogenic response. Together, these findings suggest that Angpt1/Tie2 may contribute to the endothelial reparative effects of empagliflozin and dapagliflozin, potentially in parallel with AMPK-AKT-mTORC1 signaling.

SGLT2 inhibitors are widely used for glycemic control and cardiometabolic protection, with large clinical trials showing reduced cardiovascular events in high-risk T2D and nondiabetic populations,<sup>19,21</sup> supporting their inclusion in American Heart Association/American College of Cardiology heart failure guidelines.<sup>63</sup> However, the CANVAS/CANVAS-R trial (Canagliflozin Cardiovascular Assessment Study/Canagliflozin Cardiovascular Assessment Study–Renal) reported increased lower limb amputations, including patients without preexisting peripheral vascular disease.<sup>64,65</sup> Although the mechanism remains unclear, preclinical studies have reported mixed vascular effects of canagliflozin, including impaired recovery from ischemia<sup>66</sup> and endothelial dysfunction,<sup>26</sup> alongside studies showing reduced oxidative stress,<sup>67</sup> inflammation,<sup>47,68</sup> and barrier permeability.<sup>67,69</sup> Our data demonstrate that local canagliflozin administration impairs EC angiogenic responses without altering glycemic control (Figure S2), supporting a direct vascular effect. Local delivery of SGLT2 inhibitors may therefore reduce systemic exposure and adverse effects, particularly in elderly individuals with comorbidities or when systemic therapy is contraindicated.<sup>70,71</sup> Furthermore, local delivery may overcome impaired tissue perfusion associated with metabolic dysfunction,<sup>72</sup> while

**Figure 5 Continued.** **B**, Human umbilical vein endothelial cells (HUVECs) were treated with vehicle (VEH), empagliflozin (EMPA), dapagliflozin (DAPA), or canagliflozin (CANA) under high glucose for 48 hours, and phosphorylation of AMPK, ACC, AKT, RPS6, and 4E-BP1 was detected by Western blot (n=5–7 biological replicates from 4 independent experiments). Densitometric quantifications are shown below each blot. **C** and **D**, HUVECs were treated with VEH and CANA for 48 hours under high glucose and AMPK inhibitor compound C (CompC) for 6 hours. **C**, Scratch-wound migration assay. **D**, Phosphorylation of AMPK, AKT, RPS6, and 4E-BP1 was quantified by western blot with densitometric quantifications shown on the right (n=5–6 biological replicates from 3 independent experiments). Statistical significance was determined by 1-way ANOVA with Tukey post hoc analysis. \* $P < 0.05$ , \*\* $P < 0.01$ , \*\*\* $P < 0.001$ , \*\*\*\* $P < 0.0001$ . Data are presented as mean  $\pm$  SEM.



**Figure 6. Dose reduction and AMPK (AMP-activated protein kinase) inhibition alleviate canagliflozin (CANA)-mediated suppression of wound healing and restore AKT (AKT serine/threonine kinase)/mTORC1 (mechanistic target of rapamycin complex 1) activity.**

Experimental design follows Figure 1A with modifications: db/db mice received intradermal delivery of vehicle (VEH) control, (Continued)

strategies, such as hydrogels and microneedles may enable targeted and sustained drug exposure.<sup>73,74</sup>

All 3 SGLT2 inhibitors reduced inflammation similarly (Figure 3), indicating inflammation alone is unlikely to account for the divergence in angiogenic and reparative outcomes; differential activation of fibroblast-driven remodeling markers by empagliflozin and dapagliflozin, but not canagliflozin, further supports this conclusion.

AMPK is traditionally viewed as cytoprotective during metabolic stress and plays a key role in endothelial function<sup>75,76</sup>, where its inhibition can impair cellular functions.<sup>77</sup> However, overactivation has been shown to paradoxically impair cellular functions and inhibit EC proliferation and migration.<sup>78</sup> Our mechanistic data suggest that the divergent effects of canagliflozin may be linked to its differential regulation of the AMPK, AKT, and mTORC1 downstream proteins RPS6 and 4E-BP1 phosphorylation. In accordance with our findings, several other studies identified the AMPK/mTORC1 signaling axis to be targeted by the SGLT2 inhibitors. A phospho-proteomics study of dapagliflozin<sup>40</sup> identified AMPK (PRKAA1) and mTORC1 components (Raptor and S6Kinase) as critical downstream effectors of SGLT2 inhibition, while canagliflozin and other SGLT2 inhibitors treatment was previously shown to activate AMPK in various tissues and organs.<sup>34,79</sup> In our study, empagliflozin and dapagliflozin modestly but significantly increased AMPK phosphorylation and preserved AKT signaling in ECs, whereas canagliflozin induced excessive AMPK phosphorylation and concomitant AKT suppression. To further assess the functional relevance of increased AMPK phosphorylation, we evaluated phosphorylation of ACC, a canonical downstream target of activated AMPK,<sup>80,80</sup> and observed increased ACC phosphorylation in both mouse and EC samples. These findings indicate that SGLT2 inhibitor-induced AMPK activation is functionally active and engages downstream targets. Notably, while crosstalk between AMPK and AKT signaling and their consecutive activation is essential for EC angiogenic functions under hypoxic conditions,<sup>81,82</sup> activation of AMPK can also lead to AKT inactivation in several cell types,<sup>83</sup> in agreement with our findings. These opposing effects may contribute canagliflozin-mediated inhibition of angiogenic function, and sustained AKT suppression may impair vascular repair in ischemic settings such as PAD.

Given AMPK's upstream inhibitory role on mTORC1,<sup>35</sup> we examined canonical downstream targets. Only canagliflozin reduced phosphorylation of RPS6 and 4E-BP1, consistent with selective mTORC1 suppression. mTORC1 coordinates metabolic activity, angiogenic growth, and cytoskeletal remodeling in ECs.<sup>84</sup> Although transient mTORC1 inhibition may be beneficial in some vascular contexts,<sup>85</sup> sustained suppression can impair wound healing<sup>86</sup> and limit EC angiogenic functions.<sup>84</sup> Our data therefore suggest that canagliflozin-mediated co-inhibition of AKT and mTORC1 may compromise reparative endothelial function, distinguishing its mechanism from empagliflozin and dapagliflozin and supporting endothelial mTORC1 as a regulatory node in SGLT2 inhibitor-mediated wound healing.

In further support, pharmacological AMPK inhibition using CompC decreased AMPK phosphorylation at wound sites, partially improved wound closure and perfusion, induced angiogenic markers, and removed the suppressive effect of AMPK on AKT and mTORC1 signaling. These findings underscore that uncontrolled AMPK activation may contribute to canagliflozin-associated EC angiogenic dysfunction and suggest that combining SGLT2 inhibitors with selective AMPK modulators or mTORC1 pathway agonists could enhance therapeutic efficacy while mitigating adverse vascular effects.

The pro-angiogenic effects of SGLT2 inhibitors (empagliflozin/dapagliflozin) may extend beyond wound healing to diabetic PAD, a condition associated with impaired angiogenesis, endothelial dysfunction, and diabetes-related epigenetic changes.<sup>87–89</sup> SGLT2 inhibitors have shown beneficial vascular effects in preclinical PAD models,<sup>22,39</sup> and future studies should evaluate whether empagliflozin and dapagliflozin improve outcomes in limb ischemia and critical limb-threatening ischemia in diabetic patients. Consistent with reports of impaired hindlimb repair and reduced microvessel density with canagliflozin,<sup>27</sup> our data demonstrate that empagliflozin and dapagliflozin enhance endothelial angiogenic responses under diabetic conditions through AMPK/AKT/mTORC1 signaling, supporting a potential therapeutic strategy for vascular repair.

Several limitations should be acknowledged. First, only male mice were used for in vivo studies, and sex-specific effects were not evaluated in human organoids. Male

**Figure 6 Continued.** CANA (1 or 10 mg/kg), or CANA (10 mg/kg) + compound C (CompC). **A**, Wound closure, **B** perfusion by laser speckle contrast imaging (LSCI) imaging (n=6 mice, 3 wounds/mouse), and **C** granulation tissue thickness (GTT) by H&E staining (n=6 mice/group) and **D** mRNA expression of angiogenesis-related genes by reverse transcription quantitative polymerase chain reaction assessed in mouse skin wound samples (n=6 mice/group). **E**, Phosphorylation levels of AMPK (AMP-activated protein kinase), AKT (AKT serine/threonine kinase), and mTORC1 (mechanistic target of rapamycin complex 1) pathway targets (RPS6 [ribosomal protein S6] and 4E-BP1 [eukaryotic translation initiation factor 4E-binding protein 1]) in wound tissue were measured by Western blot with densitometric quantifications shown on the right (n=6 mice, 1–2 wounds/group). Statistical significance was determined by **(A)** 2-way ANOVA (time×treatment) and remaining panels by 1-way ANOVA with Tukey post hoc analysis. \*  $P < 0.05$ , \*\*  $P < 0.01$ , \*\*\*  $P < 0.001$ , \*\*\*\*  $P < 0.0001$ . Data are presented as mean±SEM. *ANGPT1* indicates angiopoietin 1; *ANGPT2*, angiopoietin 2; *KDR*, kinase insert domain receptor; *PECAM1*, platelet and endothelial cell adhesion molecule 1; *THBS1*, thrombospondin 1; *TIE2/TEK*, TEK receptor tyrosine kinase (also known as TIE2); and *VWF*, von Willebrand factor.

db/db mice were used to maintain consistency with this established model of diabetic wound healing.<sup>90</sup> Sex is a biological variable in vascular biology and wound healing,<sup>91</sup> and future studies should assess sex-dependent effects on angiogenesis, endothelial regeneration, and barrier function. Second, the molecular pathways underlying canagliflozin's AMPK-dependent suppression of angiogenesis require further investigation, particularly whether canagliflozin-mediated AMPK activation directly regulates mTORC1 or its downstream effectors. Third, canagliflozin's broader inhibitory profile, including partial SGLT1 inhibition, may contribute to its unfavorable vascular effects. Fourth, the long-term impact of AMPK-AKT modulation on matrix remodeling and scar formation remains to be determined, particularly in chronic wounds. Finally, additional studies are needed to optimize local delivery and define pharmacokinetic parameters for SGLT2 inhibitors. In our study, local doses were substantially lower than systemic doses associated with wound healing benefits, and although reducing canagliflozin improved healing, the effect remained less pronounced than empagliflozin or dapagliflozin, suggesting that dose optimization and advanced delivery strategies may further improve efficacy.

In conclusion, empagliflozin and dapagliflozin demonstrate distinct proangiogenic and wound-healing advantages compared with canagliflozin in murine in vivo and human ex vivo models, providing new insights into the vascular effects of SGLT2 inhibitors and emphasizing their potential as therapeutic modalities in diabetic wound healing and provide the basis for future clinical studies stratifying patients by baseline microvascular health to optimize SGLT2 inhibitor selection and minimize vascular complications.

## ARTICLE INFORMATION

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### Affiliation

Department of Medicine, Molecular Cardiology Research Institute (FB, G.F.G., C.K., K.H., D.V.R., R.B., A.Z., P. Sahu, G.S.H., B.I.) and Division of Vascular Surgery, Cardiovascular Center (P. Salehi), Tufts Medical Center, Boston, MA.

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### Disclosures

None.

### Supplemental Material

Tables S1–S2  
Figures S1–S8  
Major Resources Table  
Unedited Western Blots  
ARRIVE Guidelines

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